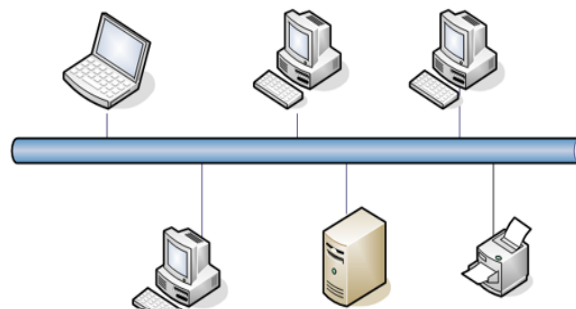


Day -7 : Network Topologies

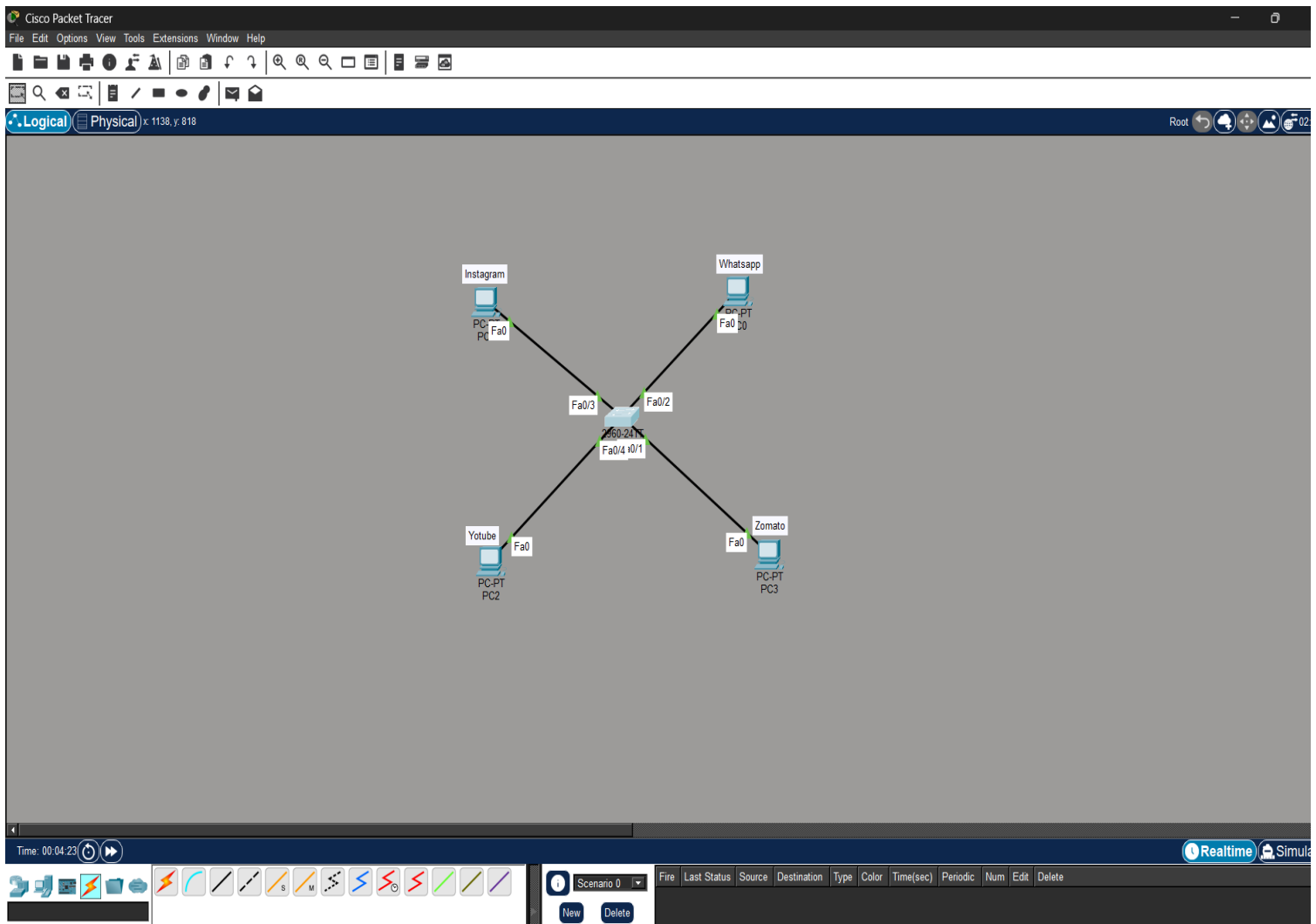
1. Image of Bus Topology :

- In the language of networking, a 'Bus' is the central cable or a single coaxial cable that connects all devices on a Local-Area Network. It is also called the backbone. This is often used to describe the main network connections constituting the internet.
- Bus network are fairly inexpensive and are easy to install therefore ideally suitable for small networks.
- Ethernet system utilize a bus Topology.

BUS Topology



2. Image of Star Topology :



3. Ring Topology :

RING Topology

- When each device is serially connected in a closed ring pattern with a single cable then the network system is called ring topology. As a train crosses all the platforms to reach its destination, ring topology does the same to pass a message from one node to another node.



4. Compare Mesh topology and Star Topology :

Topology	Good Things (Advantages)	Bad Things (Disadvantages)
Mesh Topology <i>(Every device is directly</i>	1. Super Reliable: If one wire or device breaks, the data just takes a different path. The internet won't stop working.	1. Very Expensive: It needs tons of cables and extra plug ports, which costs a lot of money.

Topology	Good Things (Advantages)	Bad Things (Disadvantages)
<i>connected to every other device)</i>	2. No Main Boss Device: There is no single central box that can break and ruin everything for everyone.	2. Hard to Set Up: Connecting every single device to each other gets messy and complicated very quickly.
Star Topology <i>(All devices plug into one central box, like a hub or switch)</i>	1. Easy to Build & Fix: It looks like a star—simple to plug in, and easy to find which wire has a problem. 2. One Device Breaking Doesn't Hurt Others: If your PC cable breaks, only your PC loses connection. Everyone else stays online.	1. Central Box Dependency: If the main central box breaks, the entire network goes completely dead. 2. Can Get Slow: Because all data has to pass through that one central box, it can slow down if too many people use it at once.